

Techno-Economic Assessment of Atmospheric Carbon Removal via Industrial Composting of Biomass Waste with CO₂ Capture

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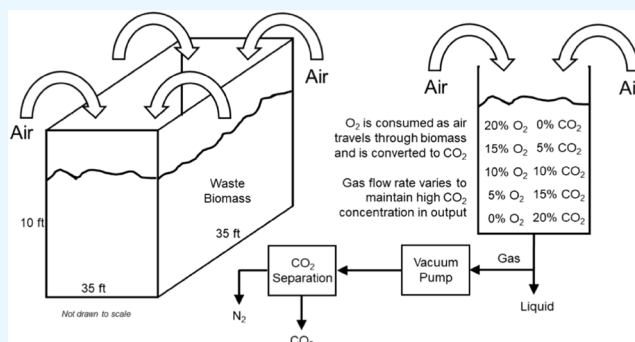


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ABSTRACT: The techno-economic feasibility of removing atmospheric CO₂ via industrial composting of high-moisture, heterogeneous biomass waste with CO₂ capture is assessed. The novel composting process generates a relatively high concentration of biogenic CO₂ for capture, while avoiding the emission of methane and generating a nutrient-rich soil amendment. Empirical data generated from compost bioreactors at 0.5 L and 25 L scales inform the techno-economic assessment. The levelized cost of CO₂ capture is estimated to be USD 85 per tonne of CO₂ (tCO₂), while the levelized cost of CO₂ removal is estimated to be USD 102/tCO₂ under base case assumptions. Sensitivity analysis indicates that a biomass feed rate and tipping fee are the most sensitive parameters. A compost facility with a biomass feed rate of 45,990 dry t-biomass/year has the potential to remove over 26,500 tCO₂/year from the atmosphere with gaseous CO₂ capture. Industrial composting of biomass waste with CO₂ capture has the potential to be a viable biomass carbon removal and storage (BiCRS) technology and warrants further investigation.



INTRODUCTION

To avoid severe consequences of climate change, emissions of greenhouse gases such as carbon dioxide (CO₂) and methane (CH₄) must not only be avoided but must also be actively removed from the atmosphere and sequestered.¹ Many pathways are being developed for the capture and removal of atmospheric CO₂, including direct air capture, mineralization, reforestation, soil carbon sequestration, and biomass carbon removal and storage (BiCRS).² BiCRS pathways rely on the upstream photosynthetic fixation of CO₂ in biomass, after which innovative processes are used to convert the biomass into valuable, carbon-negative bioproducts that remove carbon from the atmosphere over their life cycle. Prominent BiCRS technologies include gasification of biomass to hydrogen with gaseous CO₂ capture, pyrolysis of biomass with bio-oil injection, and biomass burial, all of which are best suited for dry biomass feedstocks and do not fully recycle nutrients to the soil.^{2,3} Nutrient robbery entails the stripping of soil and land of critical elements present in biomass, such as nitrogen, phosphorus, potassium, carbon, and inorganic metals, and can be a serious issue if biomass removal technologies are ineffectively designed and managed.⁴ The Food and Agriculture Organization estimates that 90% of topsoil is at risk of loss by 2050 due to practices such as nutrient robbery.⁵ There is a need for BiCRS technologies that are optimized to remove atmospheric carbon via the processing of wet, heterogeneous biomass wastes while also returning nutrients to the soil. Urban organic waste is typically wet and

heterogeneous and is most often landfilled wherein CH₄ is released upon decomposition, resulting in significant warming due to the high global warming potential (GWP) of CH₄.⁶ The US Environmental Protection Agency (EPA) determined that more than one-third of municipal solid waste in landfills is wet, heterogeneous biomass, and thus, there is significant potential for CO₂ removal via the diversion of these waste materials from landfills.⁷ Commonly reported pathways for valorizing wet biomass wastes, such as food waste and manure, include hydrothermal liquefaction (HTL) and anaerobic digestion (AD).² HTL and AD are particularly renowned for their ability to generate bioenergy in the form of liquid diesel and biogas, respectively. HTL has shown promise at the pilot scale, and subsequent economic modeling has demonstrated potential for positive financial outcomes, but the complexity of the process and lack of data beyond the pilot scale have limited the commercialization of this biomass conversion process.^{8,9} AD has seen widespread adoption and success in Europe but has faced an uphill battle in the United States due to a lack of policy support and subsequent financial risks, which resulted in

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the closure of 31% of community anaerobic digesters in the years between 1994 and 2020.^{10,11} Also, the recycling of nutrients via application of digestate to soils features challenges, including the high moisture content and potential presence of pathogens.¹² The uncertainty and complexity of these two technologies limit sustainable options for dealing with organic wet waste. Notably, the leveled costs of CO₂ removal via HTL and AD at greenfield facilities are high (USD 770–1,242 per tCO₂ removed) relative to other forms of BiCRS that use dry feedstocks.² Therefore, there is a techno-economic gap that needs to be filled by a relatively low-cost, simple, and reliable BiCRS technology for wet organic wastes. One pathway positioned to address this issue with low technological uncertainty is composting, coupled with CO₂ capture.

Industrial composting involves the aerobic degradation of biomass waste on a large scale, resulting in the production of CO₂, water, and a nutrient-rich soil amendment (compost). Composting piles in previous studies have produced CO₂ concentrations ranging from 0.05–17.7%.^{13,14} Note that the relatively high concentration of 17.7% was observed in the pore space of an open windrow after 150 days of reaction, and oxygen concentrations never decreased below 5%. However, this process has thus far been largely overlooked by the growing field of CO₂ removal, with our group recently publishing the first article on composting with CO₂ capture.¹⁵ The composting process is technologically simple compared to AD and HTL and has been utilized for thousands of years to improve crop yields and soil health.¹⁶ By coupling with CO₂ capture, wet, heterogeneous bioproduct waste is utilized to remove CO₂ from the atmosphere and return nutrients to the soil via composting with gaseous CO₂ capture.

Similar to the previously mentioned BiCRS technologies, the initial removal of CO₂ from the atmosphere takes place in fields and forests through photosynthetic fixation. The biomass that results from this process is then converted into other products or is disposed of as crop and forestry waste. After a period of temporary CO₂ removal during their use phase, the products are disposed of in common forms, such as food waste, yard waste, and manure. Composting starts with the collection and mixing of these wastes to achieve a feedstock comprised of wet, heterogeneous biomass with optimal moisture content, a carbon-to-nitrogen ratio, and particle size, to name a few parameters. Critical composting parameters are listed in Table 1.

Table 1. Critical Parameters and Their Optimal Ranges for Traditional Methods of Biomass Waste Composting^{17,31}

Parameter	Range
Initial moisture content	40–60%
Carbon to nitrogen ratio	20:1–60:1
Oxygen concentration	>5%
Temperature	45–70 °C
Pile height	6–16 ft

Microbes within the compost pile begin the process of aerobic degradation as the parameters in Table 1 are met.^{17,18} The pile's temperature will elevate as a result of microbial activity, typically taking around 2–3 days to reach thermophilic conditions of 55 °C.¹⁷ In aerated static piles, temperatures over 55 °C for 3 days inactivate over 99% of pathogens following a standard set by the USDA.^{17,19} The

active composting phase with elevated temperatures and microbial activity typically lasts between 15 and 30 days.^{20,21} Some composters opt to institute a curing phase for the compost, following the active phase, to allow for further maturation and stabilization. Once applied to land, the compost returns nutrients to the soil, provides a stable layer of organic matter to reinforce topsoil, and has been shown to increase soil carbon content, thus boosting carbon draw-down.²²

Another benefit of composting is the reduction in CH₄ production relative to that which takes place in a landfill.²³ In a landfill, anaerobic conditions are prevalent, whereas composting piles are structured with gas flow systems that actively combat the development of anaerobic pockets. CH₄ reduction potential is often cited as the primary benefit of composting.^{23,24} By avoiding the generation of CH₄, composting piles produce relatively high concentrations of biogenic CO₂. For the first time, we show how a gaseous stream consisting of 20% biogenic CO₂ can be continuously produced with moderate alterations to the traditional composting process.²⁵ As shown in Figure 1, our innovative, yet simple, design involves a negatively aerated composting bay that maintains a relatively high purity of biogenic CO₂. The negative aeration design is essential for ensuring that CO₂ emissions produced during composting are directed into the amine scrubbing system for point-source CO₂ capture rather than being released into the atmosphere as in positive aeration systems. Additionally, O₂ and CO₂ sensors are incorporated to ensure optimal aeration, maximizing the CO₂ concentration as the vacuum pump actively pulls gas through the compost pile. The CO₂ concentration of ~20% achieved by composting is significantly higher than that of air (0.04%) and postcombustion gaseous streams (4–15%), which greatly reduces the energy required for downstream CO₂ purification.¹⁵ Further, the gaseous stream from composting does not contain particulates and soot that are commonly present in postcombustion streams, thereby reducing technical complexity in gas handling.¹⁵ Heterogeneous, wet biomass waste materials are sorted, mixed, and added to a 35 ft × 35 ft × 10 ft composting bay, as shown in Figure 1.

Pipes running throughout the floor of the bay pull atmospheric air through the pile, similar to other negative aeration composting reactors.²⁶ As the gas flows downward through the pile, oxygen is consumed by the microbes, while CO₂ is produced. By the time the gas reaches the bottom of the pile, the oxygen concentration is reduced to below 1%, while the CO₂ concentration increases to approximately 20%. At a concentration of 20%, CO₂ is then separated and compressed by off-the-shelf cryogenic or amine CO₂ capture units. The CO₂ product can be permanently stored via geologic injection or mineralization or utilized in a fashion that temporarily prevents the return of the carbon to the atmosphere, such as use in beverages.²

The U.S. government sets a target to achieve a leveled cost of carbon removal at USD 100/tCO₂e by 2030.²⁷ The Internal Revenue Service (IRS) provides a tax credit worth USD 85/tCO₂ to organizations in the U.S. capturing and sequestering more than 12,500 tCO₂ per year in part to support the technology development required to achieve the previously mentioned CO₂ removal target.²⁸ There is also a smaller credit worth USD 65/tCO₂ for CO₂ that is captured and used in products such as carbonated beverages. Further, emerging voluntary CO₂ removal markets offer organizations near-term

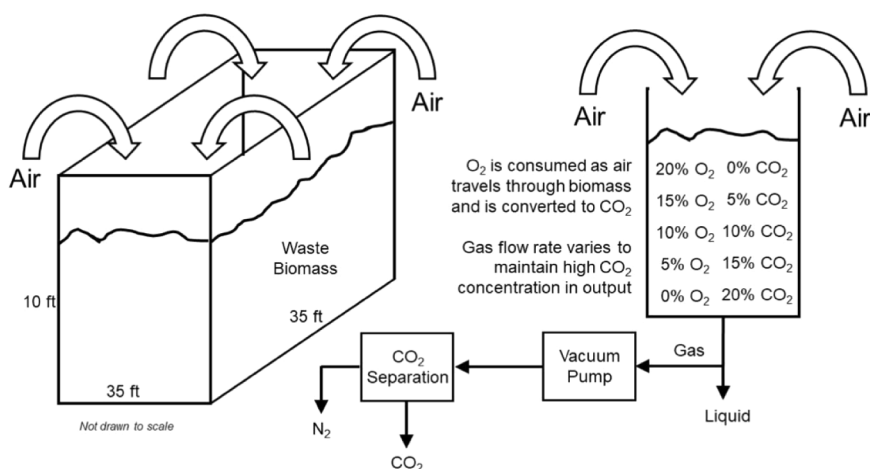


Figure 1. Simplified process flow diagram of industrial composting with the capture of gaseous carbon dioxide via negative aeration.

revenue to scale CO₂ removal technologies.²⁹ Herein, we combine experimental data, techno-economic analysis (TEA), and emissions accounting via life cycle assessment (LCA) to understand the potential commercial viability of industrial composting with CO₂ capture.

METHODS

Experimental Section. Experimental data collected from 0.5 L and 25 L composting reactors (Figure 2) provide technical details for the development of a process model that is subsequently used for TEA and LCA. Food waste, yard waste,

and agricultural waste obtained from NC State University's commercial composting facility are loaded into the bioreactors with the ideal C/N ratio and moisture content, as described in Table 1. The 0.5 L reactors are maintained at 35 °C, without agitation, and with air as the source of oxygen. At the 0.5 L scale (Figure 2A), sensors monitoring headspace gas composition are used to displace headspace gas with fresh air once the oxygen concentration decreases below 2%, at which point CO₂ concentrations are ~20%. Compost experiments last for between 15 and 30 days. Process parameters, including carbon and nitrogen content, moisture content, temperature, and mass loss, are quantified before and after each experiment. Approximately 40% of the carbon in composting feedstock is converted to CO₂ at the 0.5 L scale.¹⁵ Complete details on the methods and results from 0.5 L composting experiments with CO₂ capture can be found in a prior article published by our group.¹⁵ The 0.5 L bioreactors are closed with semicontinuous flow of gas, making scale-up challenging. An improved reactor design at the 25 L scale has been implemented wherein a continuous stream of air is drawn down through the compost, as illustrated in Figure 1 and shown in Figure 2B. The scaled-up reactor is a 25 L vertical tube wrapped in heating mats and insulation, which keeps the temperature around 55 °C in the middle of the composting pile. By continuously flowing gas at the 25 L scale as opposed to intermittently injecting gas at the 0.5 L scale, a constant stream of ~20% CO₂ is generated, while 29% of the carbon initially in the composting feedstock is converted to CO₂ over a period of 30 days at the 25 L scale. There is potential to increase the efficiency at larger scales, with additional oxygen sensors located throughout the composting pile and a more robust aeration system. See Supporting Information for additional experimental data from the 25 L bioreactor. This novel reactor design demonstrates a significant advancement in controlled composting processes by enabling continuous aeration, which optimizes the CO₂ generation and enhances the carbon conversion efficiency. The design proposed in Figure 1 is an improved and scaled-up version of the 25 L reactor, similar in structure to standard aerated commercial composting bays, with a unique emphasis placed on maximizing CO₂ production from the pile.^{30,31}

Process Model Development and Technical Assumptions. A process model is developed for composting with the CO₂ capture process using a combination of laboratory data,

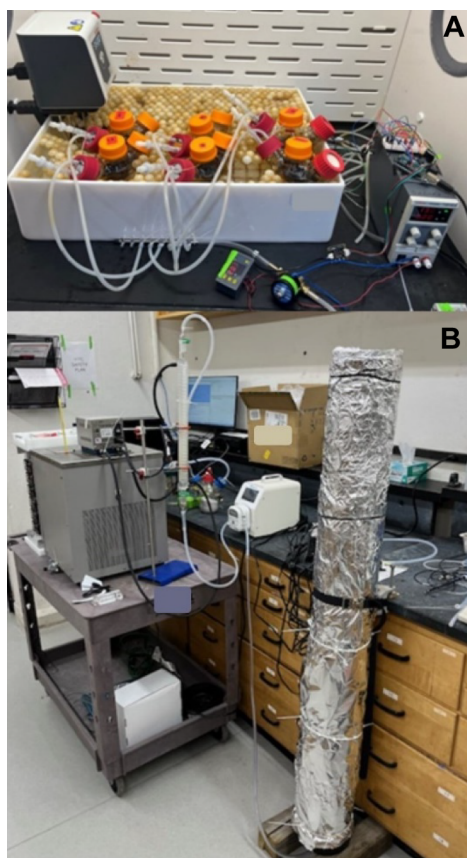


Figure 2. 0.5 L (A) and 25 L (B) compost bioreactors used to generate experimental data to inform the TEA and LCA.

literature data, and industry standards. Microsoft Excel is used for modeling the process, accounting for all mass and energy flows; Excel is deemed acceptable given the relatively straightforward process and robust data available.³² Metric tons (tonnes) are used in the reporting of all quantities that involve tons. Pertinent process model inputs and technical assumptions are listed in Table 2. The base case design criteria are largely guided by the need to capture at least 12,500 tonnes of CO₂ per year to be eligible for the 45Q tax credit in the U.S.³³

Table 2. Process Model Inputs and Technical Assumptions

Parameter	Base Case Values
Size of composting bay	35 ft L × 35 ft W × 5 ft H ²⁶
Biomass feed rate	140 dry tonnes per day
Moisture content of the feedstock	60% ¹⁷
Carbon: nitrogen ratio of feedstock	30:1 ¹⁶
Carbon conversion efficiency (CCE)	40%
Indirect emission from electricity consumption	400 kgCO ₂ /MWh ³⁴
CO ₂ capture rate	95% ²
Carbon in compost product emitted as CO ₂ upon degradation in soil	100%
Time of reaction	15 days
Concentration of CO ₂ output from composting	20 vol %
Emissions from transportation of biomass	88 gCO ₂ /t-mi ³⁵
CO ₂ capture electricity consumption	0.064 kW/kgCO ₂ /h ³⁵
CO ₂ drying and compression electricity consumption	0.073 kW/kgCO ₂ /h ³⁵
Indirect emissions from construction of site	0.21 kgCO ₂ e/USD spent in construction ³⁶
Indirect emissions from equipment materials	0.32 kgCO ₂ e/USD spent on equipment ³⁶

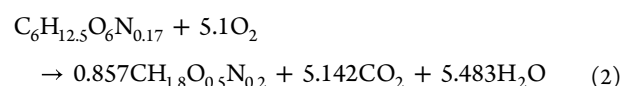
Carbon conversion efficiency (CCE) is a critical parameter defined as the carbon in biomass feedstock converted to CO₂ during the composting process (eq 1); experimental data have demonstrated CCE values ranging from 30% to 62%.

Carbon conversion efficiency

$$\begin{aligned}
 &= \text{CCE} \\
 &= \frac{\text{mass of carbon in feedstock converted to CO}_2}{\text{initial mass of carbon in feedstock}} \quad (1)
 \end{aligned}$$

The time of reaction is based on experimental results, as the bulk of microbial reactions are complete within 15 days. The concentration of outgas is based on experimental results, where CO₂ output averages 20%. All other technical assumptions are based on industry and modeling standards. The size of the base case composting bay is calculated according to desired CO₂ and compost production while adhering to industry standards.³⁷ The biomass feed rate is also determined by desired CO₂ targets and is kept within industry standards for a large composting site.³⁸ Moisture content and the carbon:nitrogen ratio of feedstock are set based on standard best composting practices.^{17,18} It is assumed that the entire composting facility, including CO₂ capture, is powered by electrical energy, and the indirect CO₂ emitted per MWh is based on literature values.³⁴ All carbon in compost is conservatively assumed to be emitted as CO₂ after field application in the LCA. There is evidence to suggest that applying compost to land can increase soil carbon stocks, but uncertainty surrounding measuring, reporting, and verifying (MRV) soil carbon sequestration led to a conservative assumption of no long-term carbon storage in the soil after compost application.³⁹ Indirect emissions from the construction of the facility and various equipment installed at the facility are accounted for using literature values.³⁶

Composting Chemistry. The biochemistry involved in converting biomass waste to CO₂ and compost is based on a simplified equation derived from results produced at the lab scale (eq 2).



The feedstock mixture consumes oxygen (in the form of gaseous air) to produce microbial biomass, CO₂, and water.

Life Cycle Assessment. The LCA accounts for both direct and indirect emissions. The Greenhouse Gas Protocol classifies direct emissions as those released from sources owned or controlled by the composting facility, while indirect emissions are those not directly released from the composting facility but still emitted due to the activities of the facility.⁴⁰ Herein, direct emissions include biomass transport, process emissions on the site, and decomposition of compost in soil. Indirect emissions include site construction and power consumption. Postconsumer organic waste is assumed to be the primary biomass waste feedstock and thus does not account for any embodied emissions in the LCA.² The emissions associated with the

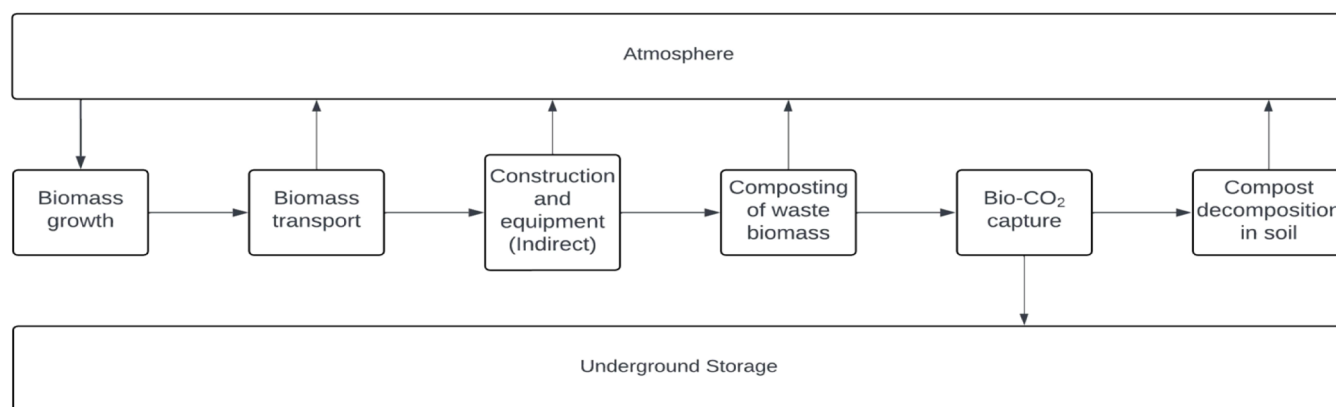


Figure 3. Sinks and sources of carbon emissions accounted throughout the life cycle of the composting with CO₂ capture and a sequestration system.

transport of finished compost products are not accounted for due to uncertainty about whether the compost would be transported at all, picked up on-site, or used in the immediate surrounding area. Figure 3 shows the various flows of CO₂e that are accounted for in the LCA. The amount of CO₂ drawn down by biomass during plant growth is defined as biomass carbon fixation (eq 3) and equals the theoretical maximum amount of removal of CO₂ from the atmosphere; eq 4 shows the CO₂ removal efficiency, which equals the actual removal of CO₂ divided by the theoretical removal of CO₂ or biomass carbon fixation. Life cycle impacts other than global warming were not considered since the inclusion of CO₂ capture to industrial composting is not expected to have significant negative impacts.

$$\begin{aligned} \text{Biomass carbon fixation} & \left(\frac{\text{tCO}_2}{\text{y}} \right) \\ &= \text{theoretical CO}_2 \text{ removal} \left(\frac{\text{tCO}_2}{\text{y}} \right) \\ &= \text{feedstock (dry t/y)} \times \\ & \quad \% \text{ C in biomass (\%)} \times \left(\frac{\text{molar mass of CO}_2}{\text{molar mass of C}} \right) \end{aligned} \quad (3)$$

$$\begin{aligned} \text{CO}_2 \text{ removal efficiency} \\ &= \frac{\text{actual CO}_2 \text{ removal accounting for life cycle emission}}{\text{theoretical CO}_2 \text{ removal}} \end{aligned} \quad (4)$$

Economic Analysis. The techno-economic assessment method for this process follows the same approach as previous studies from our group that evaluated the levelized cost of CO₂ capture and removal.^{2,41,42} The technical information from the process model is fed into an economic model to quantify the levelized costs of CO₂ capture and CO₂ removal. Pertinent economic assumptions are provided in Table 3.

The tipping fee is the amount of money that a composting site is paid to receive and process waste products and differs by region; a conservative tipping fee of USD 52/tonne is used here, which is 10% lower than the national average of USD 58/tonne.⁴³ Waste sorting equipment is included in the techno-economic model to enable the effective processing of heterogeneous organic waste. The CO₂ one-way transport distance is set to 0 miles because the site is assumed to be colocated with suitable geology for on-site injection of CO₂. A conservative national average price of USD 20/cubic yard is used for the compost selling price.⁴³ Industry standards and literature review inform the setting of plant utilization, capital scaling factor, indirect capital costs, capital cost contingency, interest rate, project life, and capital recovery factor.^{2,45,46} The capital costs associated with the CO₂ capture, drying, and compression unit operations are calculated according to eqs 5 and 6 as follows:³⁵

$$\begin{aligned} \text{CO}_2 \text{ capture capital cost (million USD)} \\ &= 233.92 \times \left(\frac{\text{tCO}_2}{\text{day}} \right)^{0.61} \end{aligned} \quad (5)$$

$$\begin{aligned} \text{CO}_2 \text{ drying and compression capital cost (million USD)} \\ &= 30.38 \times \left(\frac{\text{tCO}_2}{\text{day}} \right)^{0.77} \end{aligned} \quad (6)$$

The levelized costs of CO₂ capture and removal are calculated using a series of eqs 7 and 14 shown below. Inputs for these calculations include the capital recovery factor, biorefinery utilization, return on equity, capital costs, and revenue from tipping fees and compost sales.

Table 3. Economic Assumptions

Parameter	Base Case Value
Tipping fee	USD 52/wet tonne ⁴³
Electricity cost	USD 0.09/kWh ⁴⁴
Biomass one-way transport	37 miles
CO ₂ one-way transport	0 miles
Compost selling price	USD 20/yard ⁴³
Plant utilization	90% ²
Capital scaling factor	0.7 ²
Indirect capital costs	40% ⁴⁵
Annual maintenance costs	3% of total capital costs ⁴²
Capital cost contingency	7.5% of total capital cost ²
Percent debt financing	0%
Percent equity financing	100%
Return on equity	10% ²
Project life	20 years ²
Capital recovery factor	11.7% ⁴⁶
Cost year	2023

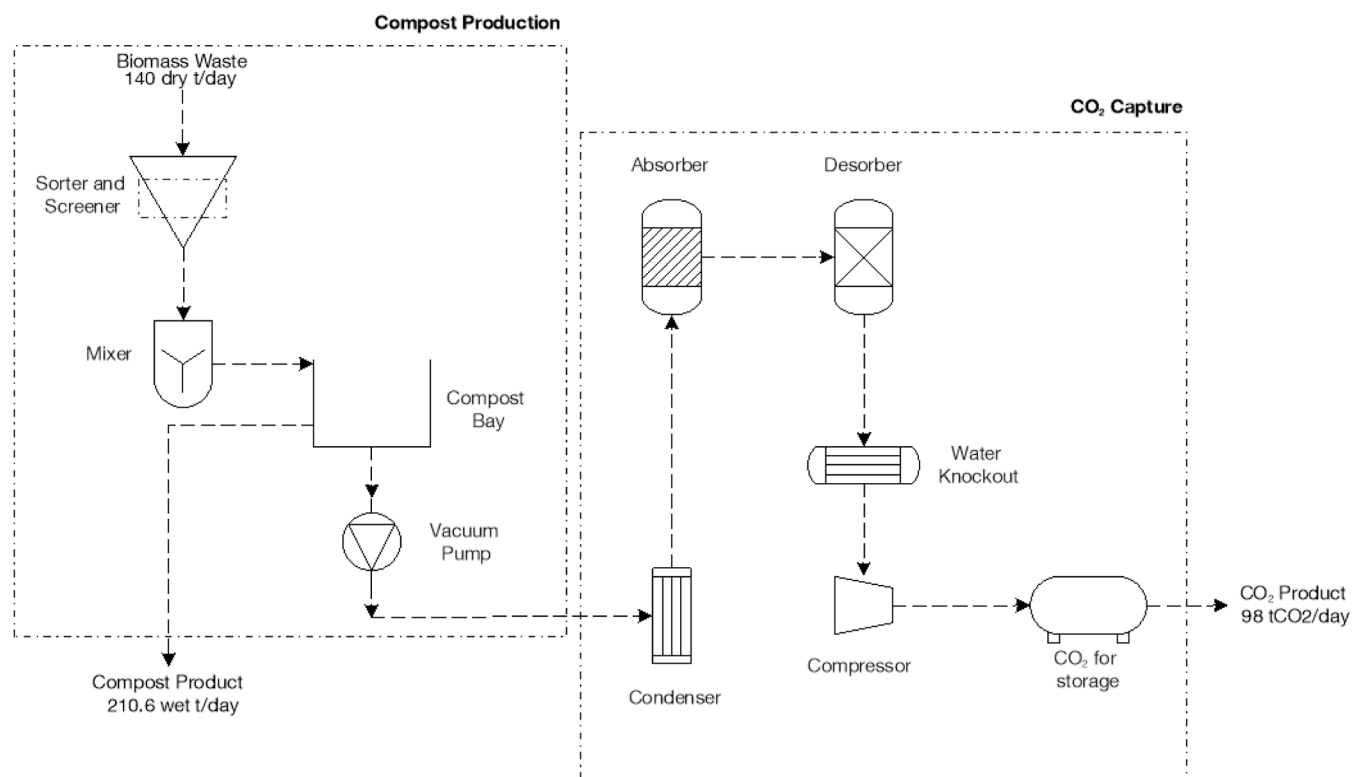


Figure 4. Process flow diagram of industrial compost production with CO₂ capture.

$$\begin{aligned} & \text{Levelized cost of CO}_2 \text{ capture} \left(\frac{\text{USD}}{\text{tCO}_2} \right) \\ &= \frac{\text{net operating cost (USD)} + \text{levelized capital cost (USD)} - \text{revenue from compost sales and tipping fee (USD)}}{\text{CO}_2 \text{ capacity (tCO}_2\text{)}} \end{aligned} \quad (7)$$

$$\begin{aligned} & \text{Levelized capital cost} \left(\frac{\text{USD}}{\text{tCO}_2} \right) \\ &= \text{capital intensity} \left(\frac{\text{USD}}{\text{tCO}_2} \right) \times \frac{\text{capital recovery factor}}{\text{biorefinery utilization}} \end{aligned} \quad (9)$$

$$\begin{aligned} \text{Revenue (USD)} &= \left(\text{tipping fee} \left(\frac{\text{USD}}{\text{t}} \right) \times \text{biomass waste (t)} \right) + \left(\text{compost produced (yd}^3\text{)} \times \right. \\ & \quad \left. \text{compost sale price} \left(\frac{\text{USD}}{\text{yd}^3} \right) \right) \end{aligned} \quad (8)$$

$$\text{Capital intensity} \left(\frac{\text{USD}}{\text{tCO}_2} \right) = \frac{\text{total capital cost (USD)}}{\text{CO}_2 \text{ capacity (tCO}_2\text{)}} \quad (10)$$

$$\begin{aligned} & \text{Total capital cost (USD)} \\ &= \text{direct capital cost (USD)} + \text{indirect capital cost (USD)} + \text{contingency cost (USD)} \end{aligned} \quad (11)$$

Direct capital cost (USD)

$$= \sum_i \text{reference cost of equipment or material}_i \text{ (USD)} \times \left| \frac{\text{cost year index}_i}{\text{reference index}_i} \right| \times \left| \frac{\text{desired capacity of equipment}}{\text{base capacity of equipment}} \right|^{\text{scaling factor}} \times \text{installation factor} \quad (12)$$

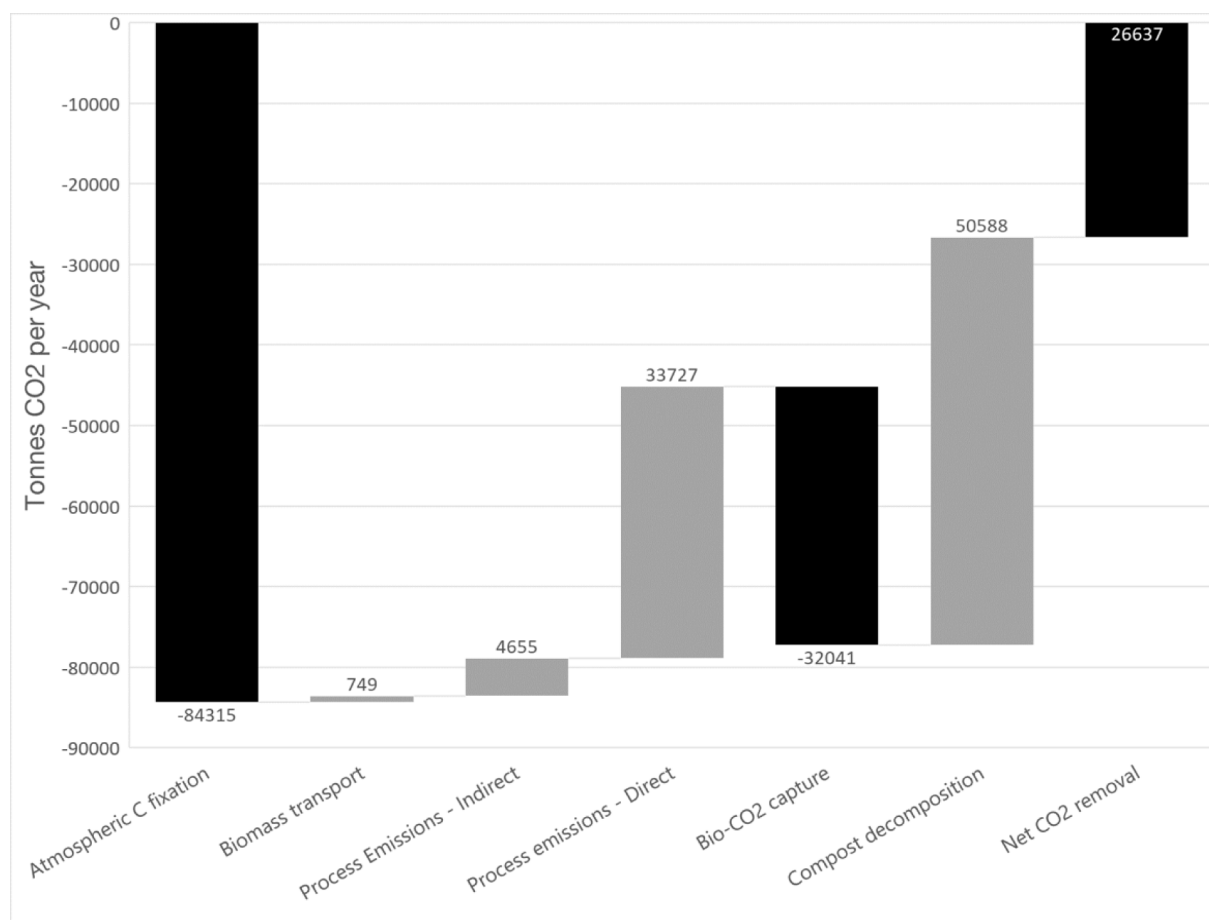


Figure 5. Waterfall plot showing the emissions and removal of carbon throughout the life cycle of the composting with the CO₂ capture and sequestration system.

Capital recovery factor (%)

$$= \frac{\text{weighted average cost of capital } (i) (\%) \times (i + 1)^{\text{project life (years)}}}{(1 + i)^{\text{project life (years)}} - 1} \quad (13)$$

Weighted average cost of capital (i) (%)

$$= (\text{return on equity } (\%) \times \text{equity financing } (\%)) + (\text{interest on debt } (\%) \times \text{debt financing } (\%)) \quad (14)$$

The cost of CO₂ removal is also calculated using eqs 7–14. The capture and removal costs differ based on the CO₂ capacities or the amount of CO₂ that can be removed or captured by the designed system. In this model, the CO₂ capture capacity equals the total CO₂ that is physically captured on-site. Meanwhile, the CO₂ removal capacity equals the amount of CO₂ removed from the atmosphere as a sum of the emission sinks and sources throughout the life cycle of the composting with the CO₂ capture system (Figure 3). CO₂ removal capacity is lower than the CO₂ capture capacity, thus increasing the levelized cost of CO₂ removal compared to that of CO₂ capture.

Process Design. The base case process design is shown in Figure 4. Biomass waste (140 dry t/day) is transported to the composting site and then sorted, screened, and mixed to achieve a carbon:nitrogen ratio of 30:1 and a moisture content of 60%. After being mixed, the feedstock is loaded into a bay

with a front-end loader. Two composting bays are emptied and loaded each day using front-end loaders. A small portion of finished compost is left in the bay to serve as an inoculum for the next composting reaction. A vacuum pump pulls gas through each pile and into the CO₂ capture system. The CO₂ capture system requires the highest energy input (43.9% of total energy consumption) of the entire process due to the multistep capture, dehydration, and compression required to make CO₂ suitable for underground injection. The final CO₂ product is of pipeline grade with a purity >99% and pressure >100 bar. The 31 composting bays generate 103 tCO₂/day of which the CO₂ processing system will capture and compress 98 tCO₂/day for subsequent storage. The energy consumption of the base case composting site is 1.3 MW, and further details on the energy consumption can be found in the Supporting information.

RESULTS AND DISCUSSION

Life Cycle Analysis. Life cycle emissions are analyzed to determine the net CO₂ removal for the process of composting with CO₂ capture. Figure 5 displays a waterfall plot wherein the black bars represent CO₂ drawdown and the gray bars represent CO₂ emissions. The far-right black bar represents the net CO₂ removal, equal to the sum of the positive and negative flows. The two steps that result in the highest CO₂ emissions are the decomposition of compost after application and the direct emissions released during the composting process. Direct process emissions include the release of CO₂ during

aerobic degradation before CO₂ capture takes place, and indirect emissions include those associated with site construction, biomass transport, electricity consumption, and compost decomposition. The production of CH₄ and nitrous oxide (N₂O) can significantly influence the direct emissions of the composting process. While these gases were not detected at the 500 mL scale, one measurement at the 25 L scale indicated CH₄ at 9 ppm and N₂O at 14 ppm, corresponding to a 46% increase in CO₂-equivalent emissions. These findings highlight the importance of optimal aeration and continuous monitoring in minimizing CH₄ and N₂O emissions. However, it is important to note that these emissions do not affect the leveled cost of CO₂ capture, as this metric strictly accounts for the operational and capital expenses associated with separating and collecting CO₂. In contrast, CH₄ and N₂O emissions could influence the leveled cost of CO₂ removal, which considers the net climate impact of the process. Since our results suggest that proper aeration can largely mitigate CH₄ and N₂O emissions and because industrial composting systems are designed to suppress these gases, we chose not to include them in our life cycle analysis. Nevertheless, future work should incorporate continuous monitoring of these emissions in scaled-up systems and assess whether additional mitigation strategies are necessary to optimize the level of CO₂ removal. In the base case scenario, 45,990 tonnes of biomass are processed in a year, resulting in the removal of 26,637 tonnes of CO₂ from the atmosphere. Thus, composting with CO₂ capture removes 0.58 tonnes of CO₂ per tonne of dry biomass. This is favorable compared to other pathways for handling wet waste, including AD and HTL, which remove 0.04–0.40 and 0.49–0.62 tonnes of CO₂ per tonne biomass, respectively.² Accounting for methane avoidance, composting 45,990 tonnes of waste instead of landfilling could also avoid between 20,000–75,000 tCO₂eq depending on the gas recovery potential of the landfill.^{47,48} The carbon removal efficiency (eq 4) of the composting with the CO₂ capture system was determined to be 32% for the base case scenario.

Capital and Operating Costs. Table 4 shows the capital and annual operating costs associated with the development of a greenfield composting site capable of CO₂ capture under base-case conditions. Figure 6 shows a percentage breakdown of the capital costs.

The CO₂ capture, drying, and compression system makes up 54% of the total capital costs, largely due to the 0.61 scaling factor used in eq 5. The modeled composting process is of relatively small scale due to limitations in the supply of wet, heterogeneous biomass and long compost reaction times, thereby making the capital cost of the CO₂ capture equipment relatively high. Traditionally, economies of scale have been difficult to achieve with industrial composting, but this notion is expected to change as incentives for composting and biogenic CO₂ capture grow.⁴⁹ Further, advancements are being made in small-scale CO₂ capture and the scaling factor will likely become closer to linear, thereby reducing the cost of CO₂ capture at composting sites in the future.⁵⁰ The biomass handling equipment and the materials for 31 composting bays make up 6% and 5% of the total capital costs, respectively.

Maintenance accounts for 35% of the annual operating costs, largely due to the CO₂ capture equipment. Labor, biomass transport, and energy costs make up 24%, 22%, and 19% of the operating costs, respectively.

Leveled Costs and Quantities of CO₂ Captured and Removed. The base case leveled cost of CO₂ capture is

Table 4. Itemized Capital and Annual Operating Costs for the Base Case Composting Facility

Itemized capital costs	
Component	Installed Cost
Composting bays include concrete, sump pump, piping, sensors, software, safety measures, actuators, valves, misc.	USD 2,787,860
Biomass handling includes waste sorting, front-end loaders, mixing, and milling	USD 3,376,639
CO ₂ capture + drying + compression	USD 28,625,131
Gas flow system	USD 310,594
Indirect cost	USD 14,040,090
Contingency	USD 3,680,610
Total includes indirect and contingency costs	USD 52,820,924
Itemized annual operating costs	
Labor	USD 1,080,000
Biomass transport	USD 1,014,597
Maintenance	USD 1,584,628
Energy	USD 899,081
Total	USD 4,578,305

USD 85/tCO₂, while the leveled cost of CO₂ removal is USD 102/tCO₂. Table 5 shows the annual costs, revenue streams, and CO₂ removal achieved in the base case scenario. Over the 20-year project period, the capital costs account for 61% of the annual costs, while operating costs make up the remaining 39%. Compost sales, tipping fees, and CO₂ sales generate revenue for the site. On average, 80% of revenue comes from tipping fees at existing, large composting sites.³⁸ In the facility modeled here, revenue from tipping fees makes up 52% of the total revenue due to the added revenue stream of CO₂. The process is a net carbon sink with 26,637 tCO₂ removed from the atmosphere per year. While the CO₂ capture system will remove a sizable amount of CO₂, the amount of avoided CO₂ is also noteworthy. The base case system of composting with gaseous CO₂ capture results in over 300,000 tCO₂e emissions being avoided by diverting organic waste from landfills, preventing anaerobic conditions and the emission of CH₄, and capturing CO₂ emitted from this waste.

Sensitivity Analysis. Twelve parameters are assessed under base, optimistic, and pessimistic cases as part of sensitivity analysis to determine which variables most impact the leveled costs of CO₂ capture, CO₂ removal, and compost production, as shown in Figures 7–9. For example, the base case compost selling price is USD 20/yd.³ In the pessimistic case, this value is decreased to USD 0/yd³ of compost sold, which drives up the cost of the whole process due to decreased revenue. For the optimistic case, this value is increased to USD 30/yd³ of compost sold, which drives the cost of the process down due to increased revenue. The 12 different parameters selected for sensitivity analysis include the biomass feed rate, return on equity, project term, tipping fee, compost selling price, CO₂ one-way transport, biomass one-way transport, electricity cost, CO₂ capture capital cost, CO₂ capture electricity consumption, CO₂ emitted per MWh, and carbon

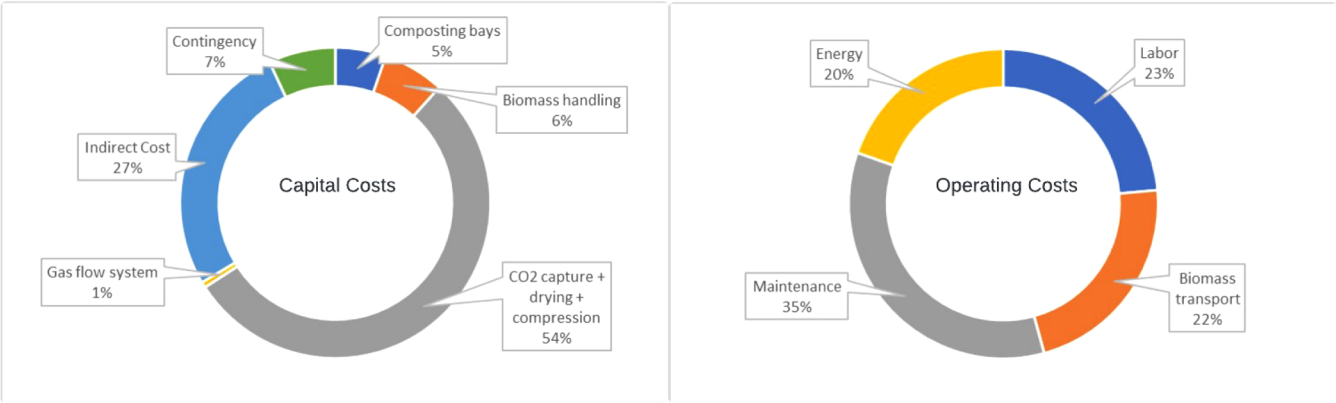


Figure 6. Percentage breakdown of capital and annual operating costs for the base case composting facility.

Table 5. Summary of Costs, Revenue, and Emission Impact for the Base Case Composting Facility. Parentheses Indicate Revenue

Expenditure or revenue	Annual cash flow
Capital	USD 6,893,695
Operating	USD 4,578,305
Compost sales	(USD 2,766,782)
Tipping fee	(USD 4,578,305)
CO ₂ sales	(USD 5,978,700)
Levelized costs	USD per tonne CO ₂
CO ₂ capture	USD 85
CO ₂ removal	USD 102
Emission Impact	Annual CO ₂ capacity
CO ₂ capture	32,041 tCO ₂
CO ₂ removal	26,637 tCO ₂
CO _{2e} avoided	306,699 tCO _{2e}

emitted from compost after being applied to soils. The biomass feed rate is varied to assess the significance of the nonlinear capital cost scaling factor, which is particularly important for industrial composting given the small scale of operation relative to other industrial bioprocessing technologies. The return on equity and project term are impactful parameters because they determine how much money will need to be paid to investors each year. The tipping fee and compost selling price are important parameters because they vary considerably by region. The average tipping fee in the US is USD 58/tonne of wet waste, but values are as high as USD 77/tonne on the

Pacific coast and as low as USD 44/tonne in the south.⁴³ A conservative value of USD 52/wet tonne of waste is used in the model for the tipping fee. A pessimistic case of USD 0/tonne of wet waste handled is used in the sensitivity analysis as well to evaluate a potential situation wherein the compost site accepts waste at no cost to the waste producer. Standard compost prices range from USD 12/yd³ to USD 60/yd³.⁴³ The USD 20/yd³ used in the model is a conservative estimate. CO₂ one-way transport is the distance that the captured and concentrated CO₂ will be moved before finally being sequestered or utilized. The CO₂ transport distance is set to 0 miles because the composting facility is assumed to be colocated with suitable geology for underground CO₂ injection. Biomass one-way transport represents the distance that will have to be traveled to pick up the biomass waste to be composted. We assume the composting facility is responsible for picking up the biomass waste from the source; thus, biomass transport is a cost incurred by the compost facility. Typically, there would be a combination of waste drop-off by producers and pickup by the compost facility. Electricity cost is the cost paid per kWh for electricity at the composting site, the majority of which is used by the biomass handling system and the CO₂ capture unit. The CO₂ capture capital costs are the costs of the amine-based CO₂ capture system and are included in the sensitivity analysis due to the outsized impact that they have on the levelized costs of CO₂ capture. Similarly, CO₂ capture electricity consumption is included in the sensitivity analysis as it represents the amount of electricity required by the amine system to capture the gaseous CO₂. The CO₂

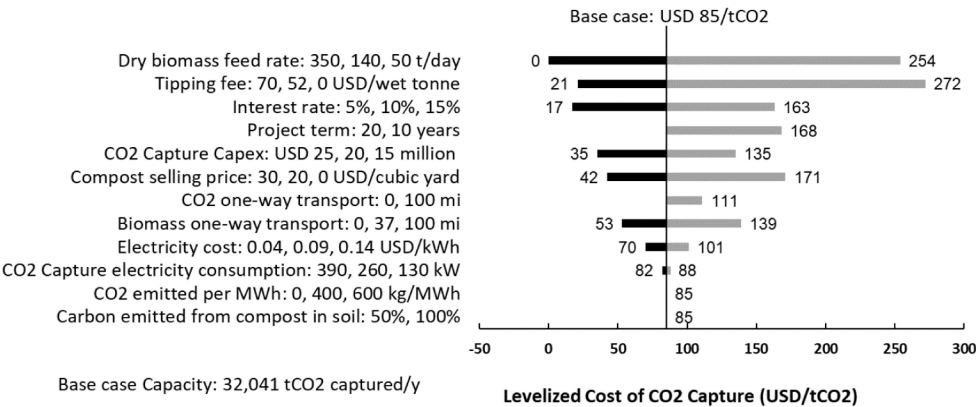


Figure 7. Tornado plot showing the sensitivity of 12 techno-economic parameters on the levelized cost of CO₂ capture.

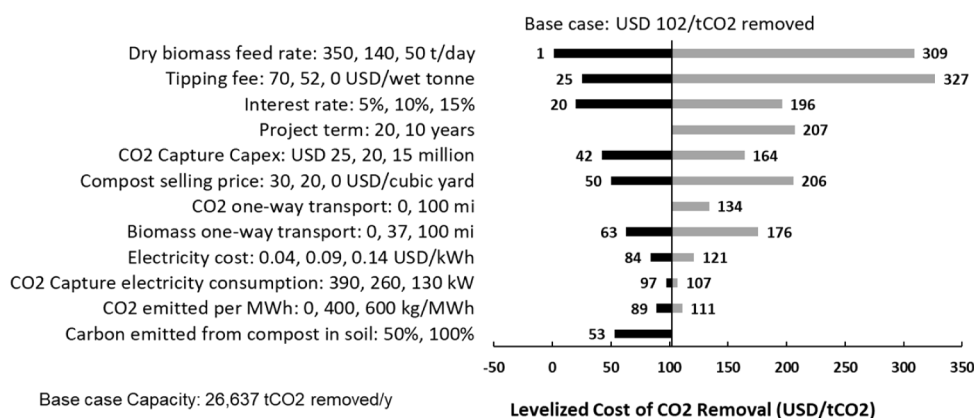


Figure 8. Tornado plot showing the sensitivity of 12 techno-economic parameters on the levelized cost of CO₂ removal.

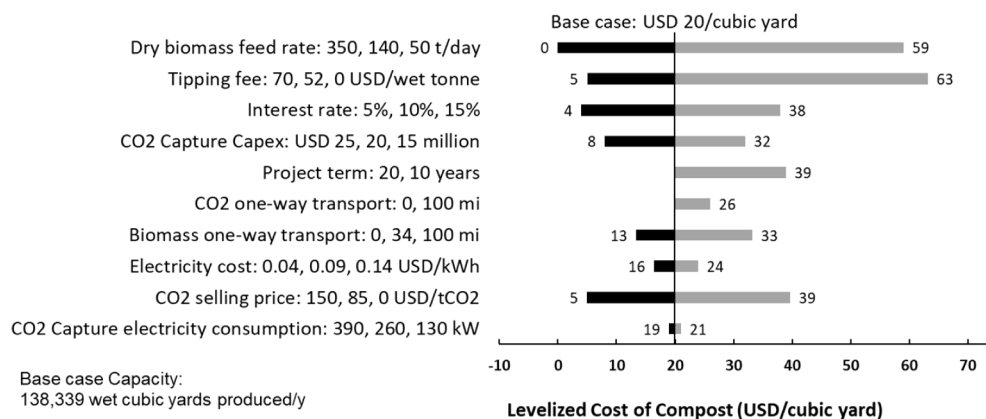


Figure 9. Tornado plot showing the sensitivity of ten techno-economic parameters on the levelized cost of compost.

emitted per MWh represents the emissions generated offsite at the local power plant. Ideally, the site could function using 100% renewable energy, thus eliminating the emissions associated with energy usage; however, this would depend heavily on the location of the site and thus is not assumed to be the case. The percentage of carbon emitted from the compost is conservatively set to 100% for the base case, as it is difficult to monitor and verify the amount of carbon that is retained by soil over a long period.⁵¹ However, the sensitivity analysis also includes the impact of reducing carbon emitted from compost to 50% in the case that MRV for soil carbon improves, and compost is shown to be a stable sink of carbon.

Figure 7 shows the sensitivity of the CO₂ capture cost to the 12 selected parameters, with costs ranging from USD 0 to USD 272/tCO₂ captured. The levelized CO₂ capture cost is equal to the annual capital and operating costs of the whole system divided by tCO₂ captured from the composting process. The cost of CO₂ capture in the base case is USD 85/tCO₂. Parameters in the base case compost process and TEA were carefully set to achieve a CO₂ capture cost of USD 85/tCO₂ to determine the feasibility of this CO₂ capture pathway when taking advantage of the 45Q tax credit in the U.S. The parameters with the highest impact on the levelized cost of capturing CO₂ are the biomass feed rate and tipping fee. Increasing the feed rate from 140 to 350 dry tonnes per day reduces the levelized cost to USD 0 per tCO₂ due to a significant reduction in capital costs incurred. Thus, at larger scales of production, the revenue from tipping fees and compost sales offsets all other costs, demonstrating the

importance of the nonlinear scaling factor for CO₂ capture equipment. Further, the tipping fee is important as it represents a major source of revenue, and when reduced from USD 52 to USD 0 per wet tonne, the levelized cost of CO₂ capture increases to USD 272 per tCO₂, representing a 120% increase. CO₂ emitted per MWh of energy consumption and the carbon emitted from compost in the soil do not impact the cost of CO₂ capture, as changes to these values affect only the cost of CO₂ removal. The levelized cost of CO₂ capture for the base case shows that it is possible to achieve USD 85/tCO₂ with the potential for reduced costs if scales of operation are increased and/or waste handling is valued higher than the national average. Conversely, the levelized cost of CO₂ capture becomes prohibitively expensive at smaller scales of operation and/or with reduced tipping fees. The capture of biogenic CO₂ via composting is more costly than the capture of biogenic CO₂ from other established industrial bioprocesses, including fermentation and pulp and paper.^{41,52} However, the costs of CO₂ removal may be more competitive due to particular advantages inherent to the composting process.²

Figure 8 shows the sensitivity of the CO₂ removal cost to the 12 selected parameters, with costs ranging from USD 1 to USD 327/tCO₂ removed. The levelized CO₂ removal cost is equal to the annual capital and operating costs of the whole system divided by the annual tCO₂ removed throughout the entire life cycle of the system. CO₂ removal accounts for GHG sinks and sources such as the CO₂ removed from the atmosphere during biomass growth and the direct and indirect GHG emissions throughout the life cycle. The cost of CO₂ removal in the base

case is USD 102/tCO₂. In comparison to the cost of CO₂ capture, the cost of CO₂ removal accounts for life cycle emissions, thereby increasing the cost. Two emission sources are included in the sensitivity analysis to understand their effects on the cost of CO₂ removal, namely, the CO₂ emitted per MWh consumed by the composting facility and the carbon emitted from compost decomposition in soil. The leveled cost of CO₂ removal decreased ~50% to USD 52 per tCO₂ under the scenario of 50% carbon in compost remaining sequestered in the soil for at least 100 years, whereas the decrease in cost due to the lower electricity emission factor was less significant, as shown in Figure 8. The biomass feed rate and tipping fee are again the most influential parameters for the cost of CO₂ removal. The costs of CO₂ removal under the most pessimistic cases range from USD 309 to USD 327 per tCO₂, which is still competitive with other forms of CO₂ removal, including direct air capture.² The composting with the CO₂ capture process has inherent advantages over other industrial bioprocessing technologies when it comes to the techno-economic feasibility of CO₂ removal. Specifically, ethanol fermentation, pulp and paper processing, and AD are three robust bioprocess technologies in the industrial bioeconomy that emit significant quantities of biogenic CO₂. Ethanol fermentation usually involves intensive farming and cultivation of corn or other starch crops as well as energy-intensive operations downstream in the supply chain, thereby making the CO₂ removal efficiency low even with CO₂ capture and use of renewable, carbon-free electricity.⁵³ Pulp and paper operations have complex and energy-intensive operations downstream in the supply chain. The cost of CO₂ removal via AD with CO₂ capture is approximately USD 280–450 per tCO₂ due to the low carbon removal efficiency.² Composting with CO₂ capture has a relatively high carbon removal efficiency due to the high conversion of biogenic carbon to CO₂, low energy intensity, and simplified operations downstream in the supply chain. There is potential for a promising synergy between AD and composting, wherein post-AD digestate is composted with CO₂ capture and then applied to land as a fertilizer. The direct application of digestate to soils poses a risk for excessive runoff and nitrogen volatilization.^{54,55} Composting digestate before application inactivates pathogens, stabilizes nutrients, and increases the amounts of nitrogen and carbon available for use by soils.

Figure 9 shows the sensitivity of compost cost to 10 selected parameters, with costs ranging from USD 0 to USD 63/cubic yard. The cost of compost in the base case is USD 20/cubic yard. The variables tested are similar to those tested in the sensitivity analyses on CO₂ capture and removal, with the exceptions being the exclusion of CO₂ emitted per MWh and the carbon emitted from compost, as neither of these impacts the cost of compost. This particular sensitivity analysis has CO₂ set as the coproduct and compost as the main product in the TEA, which differed from the prior two sensitivity analyses wherein CO₂ was the main product. Again, the biomass feed rate has the most significant impact on the total cost of compost, followed by the tipping fee and the interest rate. The range of costs is not as extreme relative to the CO₂ capture and removal analyses, in part due to the large quantity of compost being produced as opposed to the smaller amount of CO₂ captured and sold. USD 20/yd³ of compost is a relatively low-cost composting product, with costs as high as USD 60/yd³ in certain areas in the U.S.⁴³

CONCLUSIONS

Industrial composting with CO₂ capture and sequestration represents an overlooked opportunity for biomass carbon removal and storage (BiCRS). We demonstrate the techno-economic feasibility of implementing CO₂ capture and sequestration at a greenfield, large-scale composting facility at USD 85/tCO₂ captured or USD 102/tCO₂ removed, while maintaining a standard compost selling price of USD 20/yd.³ With a biomass feed rate of 140 dry t-biomass/day, 26,637 tCO₂/y could be removed from the atmosphere, representing a 32% carbon removal efficiency. In addition, over 300,000 tCO_{2e}/y would be avoided in large part due to the elimination of methane emissions that typically result from the decomposition of heterogeneous, wet organic waste in landfills. Relative to anaerobic digestion and hydrothermal liquefaction, industrial composting has advantages that make it potentially more impactful as a means to utilize heterogeneous, wet biomass waste for atmospheric carbon removal. There is potential synergy between anaerobic digestion and composting with CO₂ capture that could increase the carbon removal efficiency and thereby lower the costs. The cost of CO₂ removal via composting is most sensitive to the biomass feed rate and tipping fee, both of which should be investigated in future work. The nonlinear scaling factor used in the costing of CO₂ capture systems warrants special attention for relatively small-scale operations like industrial composting. The process of composting with CO₂ capture should be scaled up further to corroborate assumptions made in TEA and LCA.

ASSOCIATED CONTENT

Supporting Information

The Supporting Information is available free of charge at <https://pubs.acs.org/doi/10.1021/acsomega.4c10158>.

Mass balance results and O₂/CO₂ sensor readings from 25 L composting trial experiments; annual energy consumption, labor costs, and variable operating costs are also provided for the base case composting operation (PDF)

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Author Contributions

W.J.S., E.W., and J.D. conceptualized the study; E.W. generated and analyzed the experimental data; W.J.S., N.C., and E.W. developed the process model and conducted the techno-economic analysis; W.J.S., J.D., and E.W. conducted the life cycle analysis; E.W. wrote the original manuscript; W.J.S. reviewed and edited the manuscript with contributions from all coauthors.

Notes

The authors declare no competing financial interest.

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